

Experiment 8

Aim: Use Postman to test sample API.

```
Command Prompt - node ser X + v
Microsoft Windows [Version 10.0.26200.8117]
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C:\Users\Lenovo>mkdir simple-api
A subdirectory or file simple-api already exists.

C:\Users\Lenovo>cd simple-api

C:\Users\Lenovo\simple-api>npm init -y
Wrote to C:\Users\Lenovo\simple-api\package.json:

{
  "name": "simple-api",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node server.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs",
  "dependencies": {
    "express": "^5.2.1"
  },
  "devDependencies": {}
}

C:\Users\Lenovo\simple-api>npm install express

up to date, audited 66 packages in 2s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\Lenovo\simple-api>node server.js
Server running on port 3000
```

Experiment 9

Aim: Use pytest to test python application.

```
[notice] A new release or pip is available: 25.0.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\Lenovo>python -m pytest --version
pytest 9.0.2

C:\Users\Lenovo>cd PythonDemo

C:\Users\Lenovo\PythonDemo>python -m pytest -v
===== test session starts =====
platform win32 -- Python 3.11.1, pytest-9.0.2, pluggy-1.6.0 -- C:\Program Files\Python311\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Lenovo\PythonDemo
collected 10 items

test_sample.py::test_add PASSED [ 10%]
test_sample.py::test_multiply_fail FAILED [ 20%]
test_sample.py::test_divide_by_zero PASSED [ 30%]
test_sample.py::test_is_even PASSED [ 40%]
test_sample.py::test_add_param[1-2-3] PASSED [ 50%]
test_sample.py::test_add_param[2-2-4] PASSED [ 60%]
test_sample.py::test_add_param[5-5-11] FAILED [ 70%]
test_sample.py::test_factorial_normal PASSED [ 80%]
test_sample.py::test_factorial_negative PASSED [ 90%]
test_sample.py::test_factorial_large PASSED [100%]

===== FAILURES =====
_____ test_multiply_fail _____

def test_multiply_fail():
> assert pythondemo.multiply(4, 3) == 11 # ✗ wrong on purpose
      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       assert 12 == 11
E       + where 12 = <function multiply at 0x0000024E5BE61800>(4, 3)
E       +   where <function multiply at 0x0000024E5BE61800> = pythondemo.multiply

test_sample.py:11: AssertionError
_____ test_add_param[5-5-11] _____

a = 5, b = 5, result = 11

@pytest.mark.parametrize("a,b,result", [
    (1, 2, 3),
    (2, 2, 4),
    (5, 5, 11)
])
def test_add_param(a, b, result):
> assert pythondemo.add(a, b) == result
E       assert 10 == 11
E       + where 10 = <function add at 0x0000024E5BE61760>(5, 5)
E       +   where <function add at 0x0000024E5BE61760> = pythondemo.add

test_sample.py:26: AssertionError
===== short test summary info =====
FAILED test_sample.py::test_multiply_fail - assert 12 == 11
FAILED test_sample.py::test_add_param[5-5-11] - assert 10 == 11
===== 2 failed, 8 passed in 0.26s =====

C:\Users\Lenovo\PythonDemo>
```